WORKED SOLUTION

STEP III 1995, Question 13

Probability and Statistics • Noisy Binary Channel

STEP III 1995, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

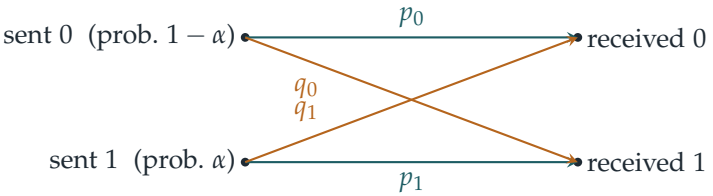
step.maths.org/questions/95-s3-q13

The key idea

Everything follows from looking at one digit at a time. A digit travels through a "binary channel": it is sent as 0 or 1, then either survives or is flipped. Once we know what happens to a single digit, the whole message is just many independent copies, so probabilities multiply. For the booster, the trick is to notice that a digit arrives correctly after two legs if it is flipped an *even* number of times.

Solution

Throughout, write $N = 10^k$ for the length of the message. We assume, as the question intends, that the digits of the original message are independent (each is a 1 with probability α), and that the cable treats each digit independently of all the others. The single-digit picture is below.



The whole message received correctly

Let C be the event that a particular digit is received correctly. The digit is either a 0 that survives or a 1 that survives, so

$$\mathrm{P}(C) = (1-\alpha)p_0 + \alpha p_1.$$

By independence across the N digits,

$$\mathrm{P}(\text{entire message correct}) = \left[(1-\alpha)p_0 + \alpha p_1\right]^{10^k}.$$

The probability β

A digit is received as a 1 either if it was a 1 and survived, or if it was a 0 and was flipped:

$$\beta = \alpha p_1 + (1-\alpha)q_0.$$

Now

$$\begin{aligned}\beta = \alpha &\iff \alpha p_1 + (1-\alpha)q_0 = \alpha \iff (1-\alpha)q_0 = \alpha(1-p_1) = \alpha q_1 \\ &\iff q_0 = \alpha(q_0+q_1) \iff \alpha = \frac{q_0}{q_0+q_1},\end{aligned}$$

where the last step divides by q_0+q_1, which is positive unless the cable is perfect ($q_0 = q_1 = 0$, in which case $\beta = \alpha$ trivially). Every step is reversible, so this is the required "if and only if". In words: the proportion of 1s is unchanged by the cable exactly when the flow $0 \to 1$ (that is, $(1-\alpha)q_0$) balances the flow $1 \to 0$ (that is, αq_1).

A message of all zeros

Suppose the condition holds and every received digit is 0. Because the pairs (sent digit, received digit) are independent across positions, the conditional probability that the whole message is correct is the product of the single-digit probabilities $\mathrm{P}(\text{sent } 0 \mid \text{received } 0)$.

> **Ask yourself** How do we turn "received 0" into a statement about what was sent?

By Bayes' theorem. The probability that a digit is received as 0 is $1-\beta$, and the condition says $\beta = \alpha$, so $1-\beta = 1-\alpha$. Hence

$$\mathrm{P}(\text{sent } 0 \mid \text{received } 0) = \frac{(1-\alpha)p_0}{1-\beta} = \frac{(1-\alpha)p_0}{1-\alpha} = p_0,$$

and therefore

$$\mathrm{P}(\text{message correct} \mid \text{all zeros received}) = p_0^{10^k}.$$

A fifty-fifty chance

With $q_0 = q_1 = q$ and $\alpha = \frac{1}{2}$, each digit is correct with probability $\frac{1}{2}(1-q) + \frac{1}{2}(1-q) = 1-q$. For a message of 10^6 digits we need

$$(1-q)^{10^6} = \tfrac{1}{2} \quad\Longrightarrow\quad 10^6 \ln(1-q) = -\ln 2.$$

Since q is tiny, $\ln(1-q) \approx -q$, giving

$$q \approx \frac{\ln 2}{10^6} \approx 6.93 \times 10^{-7}.$$

Adding a booster station

Halving the length divides q by $2^2 = 4$, since q is proportional to the square of the length. So each half of the cable flips a digit with probability $q/4$, independently of the other half (the booster just re-sends whatever it receives).

> **Ask yourself** Is "both halves transmit correctly" the only way for a digit to arrive correctly?

No: a digit that is flipped in *both* halves is flipped back. So

$$\mathrm{P}(\text{digit correct}) = \left(1 - \frac{q}{4}\right)^2 + \left(\frac{q}{4}\right)^2 = 1 - \frac{q}{2} + \frac{q^2}{8} \approx 1 - \frac{q}{2},$$

ignoring terms of order q^2. (Here the double flip is one of those terms, so it makes no difference to the answer, but it is worth seeing why.) The probability that the whole message is correct is now

$$\left(1 - \frac{q}{2}\right)^{10^6}.$$

To the same order, $(1 - \frac{q}{2})^2 \approx 1 - q$, so

$$\left(1 - \frac{q}{2}\right)^{10^6} \approx (1-q)^{10^6/2} = \left[(1-q)^{10^6}\right]^{1/2} = \left(\tfrac{1}{2}\right)^{1/2} = \frac{1}{\sqrt{2}} \approx 0.707.$$

So the booster raises the chance of a perfect message from $\frac{1}{2}$ to about 0.71. Note that we never needed the numerical value of q here: the original condition $(1-q)^{10^6} = \frac{1}{2}$ did all the work. The same idea with $e^{-10^6 q} = \frac{1}{2}$ gives $e^{-10^6 q/2} = 1/\sqrt{2}$.

Checks

- Enumerating every sent and received message of length 4 exactly (with $p_0 = 0.9$, $p_1 = 0.8$) reproduces both $[(1-\alpha)p_0 + \alpha p_1]^4$ and, under the condition on α, the conditional probability $p_0^4 = 0.6561$.
- Solving $(1-q)^{10^6} = \frac{1}{2}$ exactly gives $q = 6.9315 \times 10^{-7}$, agreeing with $\ln 2/10^6$ to five significant figures.
- With this q, the exact booster probability $\left[(1-\frac{q}{4})^2 + (\frac{q}{4})^2\right]^{10^6}$ is 0.707107, matching $1/\sqrt{2}$.
- Sense check: if $q_0 = q_1$ the condition gives $\alpha = \frac{1}{2}$, as symmetry demands.

Common slips

- Taking the error in each half to be $q/2$. The error is proportional to the *square* of the length, so it is $q/4$.

- Answering the "all zeros" part with $(1-\alpha)^N$ or p_0 alone: we need the conditional probability for each digit, raised to the power 10^k.

Beyond STEP

This is the *binary channel* of information theory. When $q_0 = q_1$ it is the binary symmetric channel, the starting point for Shannon's theory of how much information a noisy channel can carry, and for error-correcting codes, which add redundancy so that occasional flips can be detected and repaired rather than merely made rarer. The calculation with the booster is also a two-step Markov chain: the flip probabilities compose by multiplying transition matrices. Grimmett and Welsh, *Probability: An Introduction*, covers conditional probability and Markov chains at the right level.

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